Question 2: Five-class Amazon review rating prediction

BiGRU with pretrained GloVe and additive attention versus fully fine-tuned RoBERTa with masked mean pooling. Experiments use seeds 42 and 123; first-token pooling is the selected ablation. This section reports inspected saved measurements, without additional model selection after final test evaluation.

1. Data, preprocessing and controlled training

The official SetFit English split contains 200,000 training, 5,000 validation and 5,000 test reviews, balanced across stars 1-5. SetFit supplies bodies but omits titles. Titles were recovered from the original-corpus mirror using exact agreement of ID, body, language and stars, then joined as title + newline + body for both models. Null titles (33 train, one validation) remain empty. Vocabulary and preprocessing fit training only; validation chooses checkpoints. Five exact train/validation text overlaps are disclosed while retaining the required official splits. Reserved test access followed a frozen protocol.

The classic pipeline HTML-unescapes, lowercases and removes tags/URLs, then applies regex tokenization. Minimum frequency 2 yields 32,611 vocabulary entries, capped at 60,000, with PAD/UNK and a 200-token limit. GloVe 6B 100-dimensional vectors cover 87.77% of vocabulary; uncovered vectors use fixed seed 2026. RoBERTa uses original text, byte-level BPE and a 256-token limit with dynamic padding to multiples of eight. Dataset/model revisions and hashes are pinned in config.json and manifests.

Setting	BiGRU + attention	RoBERTa mean / first
Architecture	2 BiGRU layers; 128 hidden/direction; additive attention width 128	roberta-base; all encoder weights trainable; masked mean / first token
Head / parameters	Dropout 0.3; linear -> 5; 3,768,497	Dropout 0.2; linear -> 5; 124,058,885
Optimizer	Adam, learning rate 0.001	AdamW, learning rate 0.00002; decay 0.01; 8% warmup then linear decay
Batch / precision	128; float32	Global microbatch 16; accumulation 2; effective 32; recorded bfloat16
Selection / cap	Validation Macro F1; patience 2; at most 8 epochs	Validation Macro F1; patience 1; at most 3 epochs

Both use cross-entropy, gradient clipping 1.0 and minimum improvement 0.0001. GloVe is frozen for epoch one then unfrozen. Packed GRU sequences and masked attention exclude padding. RoBERTa mean pooling includes all non-padding states, including special tokens. AdamW excludes biases/LayerNorm from decay. All recorded runs used Tesla T4 hardware; optimizer/scheduler state was retained during active recovery.

Run	Seed	Epochs / best	Training h	Execution
baseline	42	7 / 5	0.22	1 GPU(s)
baseline	123	6 / 4	0.19	1 GPU(s)
roberta_mean	42	3 / 3	6.44	1 GPU(s)
roberta_mean	123	3 / 2	3.55	2 GPU(s)
roberta_first	42	3 / 3	3.56	2 GPU(s)
roberta_first	123	3 / 2	3.43	2 GPU(s)

2. Learning curves and selection

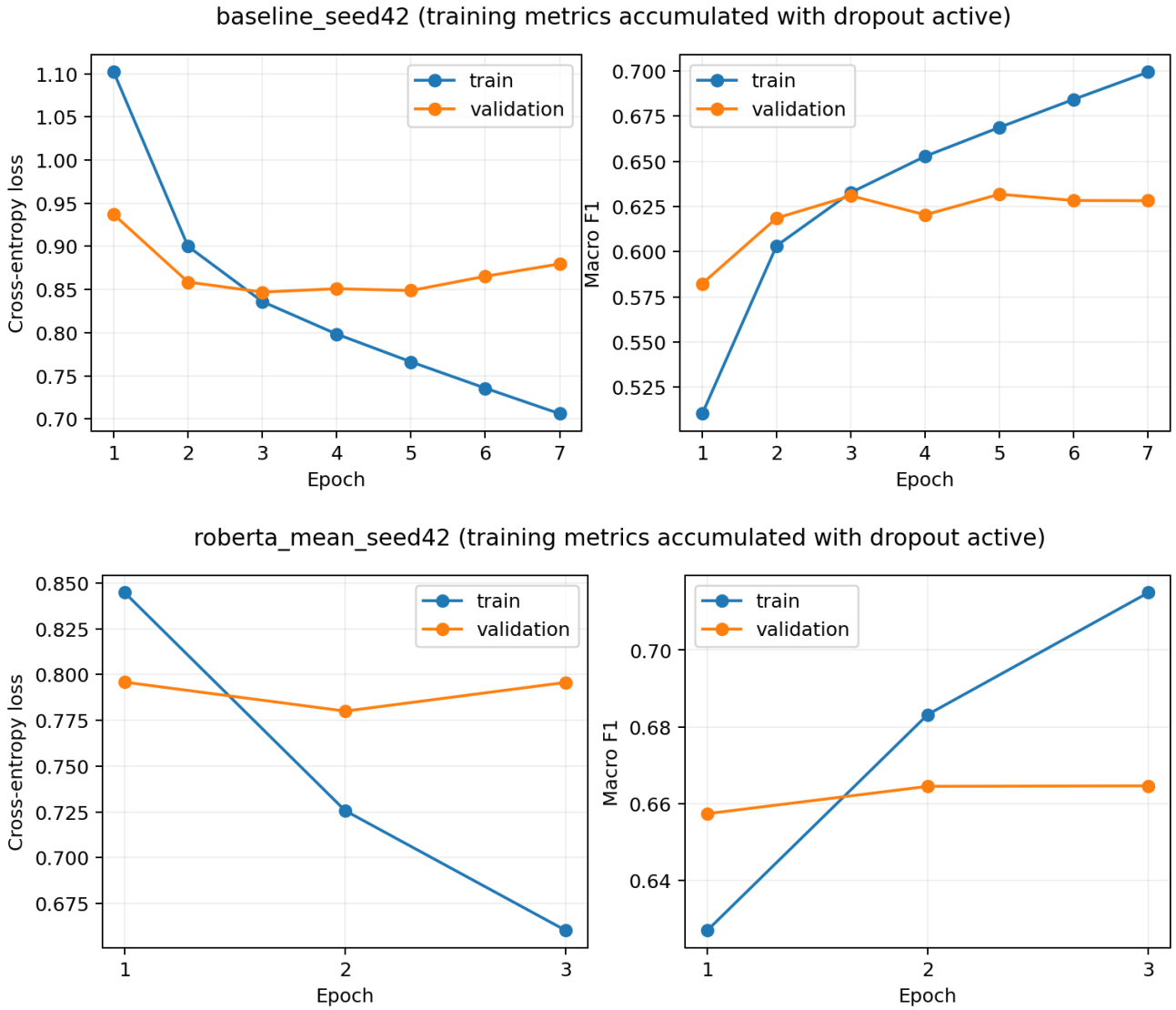


Figure 1. Training/validation loss and Macro F1 for seed 42 of the two main models. Equivalent figures for seed 123 and both ablation seeds are supplied in results/figures/. Training metrics accumulate with dropout active; validation uses evaluation mode, so the two curves are not identical evaluation protocols.

Baseline best epochs were 5/4 for seeds 42/123, with early stopping at epochs 7/6. Mean-pooling RoBERTa selected epochs 3/2; both runs completed the three-epoch cap. Checkpoints use validation Macro F1, not minimum loss or test accuracy. Full histories retain the actual losses and scores rather than only the selected epochs.

3. Frozen final test evaluation

Each main model/seed produced one final 5,000-review test prediction file. Inspection checked sample IDs, probability ranges/sums, argmax labels and recomputed metrics, including aggregate means and sample SD (ddof=1). Predictions use argmax with labels 0-4 mapped to stars 1-5. First-token models are evaluated on validation for the ablation, not additionally on test.

Model	Accuracy (%)	Macro F1	MAE
BiGRU	63.79 +/- 0.78	0.6351 +/- 0.0069	0.4277 +/- 0.0075
RoBERTa mean	67.13 +/- 0.04	0.6706 +/- 0.0010	0.3672 +/- 0.0000

RoBERTa improves accuracy by 3.34 percentage points and Macro F1 by 0.0355, while reducing MAE by 0.0605 stars. These are descriptive two-seed comparisons, not confidence intervals or significance tests. Smaller observed seed variability does not establish robustness.

True rating	BiGRU F1, mean +/- SD	RoBERTa F1, mean +/- SD
1	0.7197 +/- 0.0032	0.7454 +/- 0.0085
2	0.5483 +/- 0.0072	0.5911 +/- 0.0080
3	0.5360 +/- 0.0107	0.5792 +/- 0.0059
4	0.5908 +/- 0.0129	0.6369 +/- 0.0002
5	0.7805 +/- 0.0007	0.8004 +/- 0.0055

RoBERTa improves F1 for every rating; the largest gain is for four stars (about 0.0460). Both models identify extreme ratings more accurately than two-, three- and four-star reviews. Middle-rating wording is often mixed and neighboring rating labels can be ambiguous.

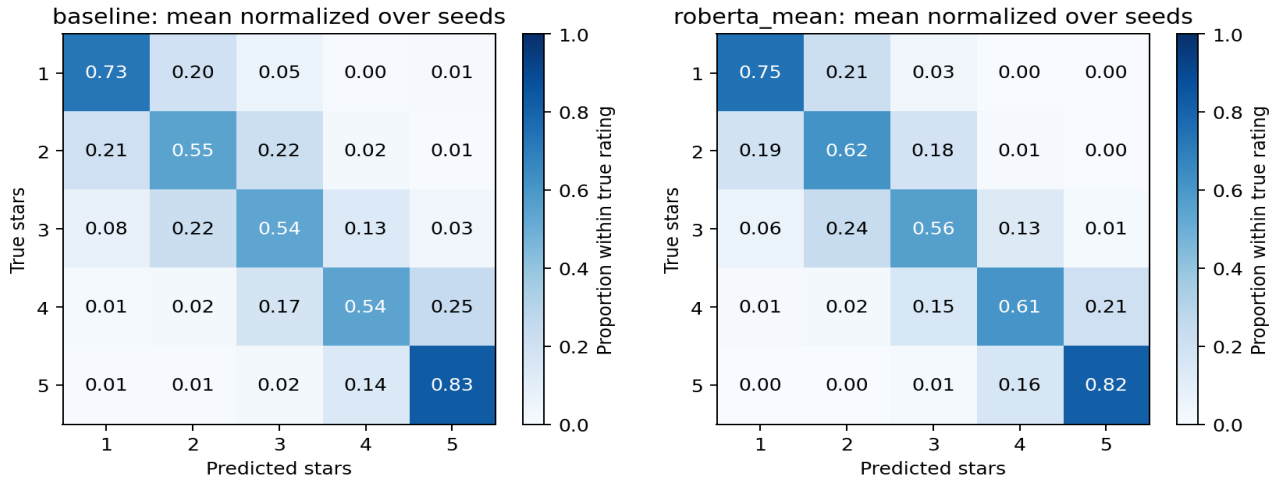
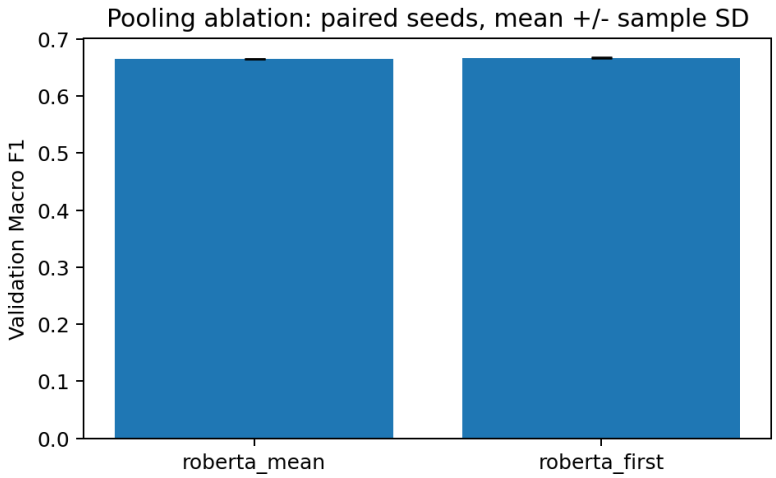


Figure 2. Row-normalized test confusion matrices averaged over seeds: BiGRU (left), RoBERTa (right). Full seed-specific matrices are supplied.

Adjacent-rating error rates fall from 30.84% to 29.49% of all reviews; distant errors (distance >=2) fall from 5.37% to 3.38%. Adjacent errors comprise 85.16% versus 89.72% of mistakes: the larger share for RoBERTa reflects fewer distant errors, not a higher overall adjacent-error rate. Extreme 1-versus-5 error rates fall from 0.29% to 0.09%.

4. Pooling ablation, reproducibility and limitations

Validation result	Mean pooling	First-token pooling
Macro F1 mean +/- SD	0.664875 +/- 0.000313	0.666848 +/- 0.000459
MAE mean +/- SD	0.3738 +/- 0.0006	0.3725 +/- 0.0016



First-token pooling is slightly better on validation in both seeds: mean-minus-first Macro F1 differences are -0.001869 (seed 42) and -0.002076 (seed 123). Its average MAE is lower by 0.0013. These small differences do not establish a universal pooling advantage. The required mean-pooling architecture remains the main final-test model; the ablation does not trigger test-driven architecture replacement.

Execution topology limits pooling-only attribution. Mean seed 42 used one GPU, while mean seed 123 and both first-token runs used two-GPU DistributedDataParallel. Global microbatch 16, accumulation 2 and effective batch 32 were preserved; gradient/communication probes passed, but rank-specific dropout streams and numerical reduction order can alter trajectories. The seed-123 pair shares topology; the seed-42 pair does not. Both baselines used one GPU. Actual topology, precision and source provenance are retained per run.

Training and evaluation were performed in the recorded Kaggle environment; CPU finalization only inspected and packaged saved files. It did not retrain or access the test corpus. Six training archives retain the selected checkpoints; the compact submission component excludes full data/caches/checkpoints. The final test protocol hash is recorded in FROZEN_PROTOCOL.json, and generated analysis can reuse its saved predictions.

Limitations include only two seeds, a single review domain, rare/OOV tokens, bounded sequence lengths, five known train/validation text overlaps, and ambiguity between neighboring star labels. RoBERTa also benefits from extensive external pretraining, so this comparison does not isolate architecture alone. No calibration, sarcasm understanding, deployment safety or out-of-domain robustness was established. Improvements should be selected on new training/validation experiments, not the final test results.

Conclusion: RoBERTa mean pooling improves the measured rating-classification and ordinal-error metrics over this GloVe-BiGRU baseline. First-token pooling has a small validation advantage in the present ablation. Neither finding establishes a general superiority across datasets, execution environments or untested hyperparameters.

5. Baseline-wrong / RoBERTa-right examples

Fixed seed 42, selected by ascending review ID: five of 500 available baseline-wrong/RoBERTa-right cases. Labels and predictions are unedited; interpretation below is a text-supported hypothesis, not a causal account of internal reasoning. Full texts are also retained in qualitative_examples.csv.

en_0002917: true 3; BiGRU 4; RoBERTa 3

Title: I also think I need to try a different coffee because I haven't been real pleased with the flavor of the coffee
Review: I've done a few so far but getting the hang of securing the foil to the capsule takes patience and practice. I've put holes in a few foils and had to redo them. I also think I need to try a different coffee because I haven't been real pleased with the flavor of the coffee. Also, the original foils on the capsules can be difficult to remove.

Mixed sentiment and practical difficulties. The review combines partial progress with flavor dissatisfaction, damaged foils and difficult capsule preparation. The neutral three-star label fits this mixed evidence. RoBERTa may better combine these qualifications; the positive four-star baseline prediction underweights complaints.

en_0003912: true 3; BiGRU 2; RoBERTa 3

Title: I won’t wear it much
Review: It’s a bit expensive and doesn’t lay right, it sticks to everything.

Rating ambiguity and malformed text. The text is strongly critical, but its recorded label is three stars. RoBERTa matches that label while the baseline predicts two. This is label agreement, not proof that three stars is the uniquely correct interpretation. The malformed apostrophes are present in the saved text.

en_0005862: true 2; BiGRU 3; RoBERTa 2

Title: Loved this purse but
Review: Loved this purse but, I can't take the bad odor coming from inside the purse. I even aired it out for two days and there was no difference. In process of returning.

Contrast and overriding complaint. The positive opening is reversed by persistent odor, unsuccessful airing and a return. RoBERTa predicts two stars while the baseline predicts three; sensitivity to the contrast and return intent is a plausible explanation, not a demonstrated reasoning process.

en_0008881: true 2; BiGRU 3; RoBERTa 2

Title: ... Expected it to be made out of a material like a bath towel
Review: Too Warm - Expected it to be made out of a material like a bath towel. Very soft, but seems like an old lady would wear this to stay warm inside a building that the A/C is on in the summertime on a 95 degree day...

Mixed attributes and unmet expectations. Softness is positive, but excessive warmth and the mismatch with expected towel-like material dominate the complaint. RoBERTa predicts two stars; the baseline predicts three. This suggests different weighting of positive attributes and dissatisfaction.

en_0009772: true 2; BiGRU 3; RoBERTa 2

Title: Could not find the answers I needed
Review: The Puzzle dictionary is too heavy to handle. The words, Clues, I tried to look up were more difficult than the clue itself. Would not suggest it for making Crossword Puzzles easier.

Negative utility and recommendation. The dictionary is described as heavy, unhelpful and difficult to use, with an explicit negative recommendation. RoBERTa matches two stars while the baseline gives three. This example concerns failure of the intended use rather than sarcasm.

6. Reverse cases and token-level importance

Two of 362 available baseline-right/RoBERTa-wrong cases, again in ascending review-ID order for seed 42. The stronger aggregate model still fails on individual reviews.

en_0000916: true 3; BiGRU 3; RoBERTa 2

Title: Nice mat but slips all over the place
Review: Doesn't stay put. Annoying.

Positive title versus short negative body. The title calls the mat nice but the body says it slips and is annoying. The baseline matches three stars; RoBERTa predicts two. The latter may overemphasize the complaint. The short mixed review illustrates uncertainty in mapping wording to exact stars.

en_0005599: true 5; BiGRU 5; RoBERTa 4

Title: Very classy and very sturdy
Review: We bought this to use in our RV. We like it very much. Very classy and very sturdy (so far). It is not light weight (ie, like Correlle) but we were not necessarily looking for anything lightweight). However, is is very attractive and so far we have enjoyed using it in our RV.

Qualification within strong approval. The reviewer repeatedly likes the product and accepts its weight as compatible with their needs. The baseline matches five stars; RoBERTa gives four. Mention of weight or the qualification so far may contribute, but the explanation remains a hypothesis.

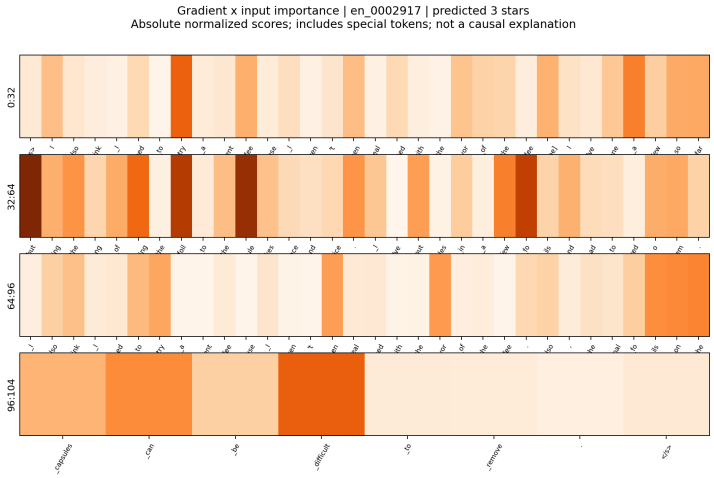


Figure 3. Absolute gradient x input importance for the predicted-class logit of the fixed RoBERTa model on en_0002917. Scores are normalized across all retained tokens, including special tokens. The displayed subset is not the complete review; full scores are in token_importance.csv.

In the saved scores, the contrast marker but (position 32) has importance 0.0398; capsule 0.0368, foil 0.0334, the fo subword of foils 0.0322, difficult 0.0264 and try 0.0261 also receive substantial weight. This sensitivity is consistent with practical difficulties and mixed sentiment, but does not establish how the model combined them or prove negation understanding. Tokens for the first occurrence of the negation have smaller scores than some context tokens. The example has 104 retained tokens, below the 256-token limit; truncation is therefore not the explanation for this particular difference. Other long reviews may lose text.

Reproduction/evidence: question_2/README.md and KAGGLE_RUN_GUIDE.md describe exact phases; config.json, manifests, environment records, training_records/, final prediction CSVs and metric JSONs retain run-level provenance. Qualitative interpretations are based on the saved review text and model predictions.

Sources: Assignment PDF pp. 5-7; SetFit/amazon_reviews_multi_en and buruzaemon original-corpus mirror (exact IDs/revisions in config.json); Stanford GloVe 6B; FacebookAI/roberta-base. Full source links and recorded settings are retained in the component README/configuration.